Compact Dual-Band Textile PIFA for 433-MHz/2.4-GHz ISM Bands

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***Abstract*—A textile planar inverted-F antenna (PIFA) is proposed. By co-designing the PIFA antenna and the ground plane, a wide matching bandwidth is achieved around 433 MHz for a relatively small electrical size ($0.202 \times 0.115 \times 0.01\ \lambda_0{}^3$). A slot within the PIFA patch enables the 2.4-GHz operating band. The measured bandwidths are 35 and 309 MHz in the lower and upper bands, respectively. The corresponding radiation efficiencies are 48% and 64%. The analysis of the effect of bending and the on-body characterization show the robust performance of the antenna.**

***Index Terms*—Dual-band antenna, planar inverted-F antenna (PIFA), textile antenna.**

I. Introduction

FLEXIBLE conformal textile antennas have attracted wide attention in relation with the development of wearable systems. A lot of antenna topologies have been studied, including microstrip patches [1], [2], planar inverted-F antennas (PIFAs) [3]–[5], substrate integrated waveguide antennas [6]–[8], and antennas based on metamaterials [2], [8]–[11]. However, when the antenna is targeting a lower operating frequency band, as, for example, the 433-MHz Industrial, Scientific, and Medical (ISM) band, all the designs suffer from a narrow bandwidth [5] or a huge dimension [1].

Recently, the relationship between the size of the ground and the bandwidth has been studied by several scholars. A design flow-chart has been proposed in [12], based on characteristic mode theory [13]. This design method has been widely used in so-called "chassis antennas" for mobile terminals [13]–[15]. The chassis or the ground is actually operating as a radiator, which can greatly increase the bandwidth of the whole antenna, since the size of the chassis is normally much larger than the basic radiator itself. The key issue is to carefully choose the location of the basic radiator in order to efficiently excite the ground [13]. Inspired by this design concept, in this letter a textile antenna is presented targeting both the 433-MHz and 2.4-GHz ISM bands. Two wide bands are obtained by using slots, not only in the patch, but also in the ground plane. The wide band of the antenna guarantees a robust performance. It can handle different environments, including relatively severe bending situations, and relatively large fabrication tolerances. To the best knowledge of the authors, this is the first work that adopts characteristic mode theory to design and optimize a wearable antenna. CST Microwave Studio [16] was used in the design.

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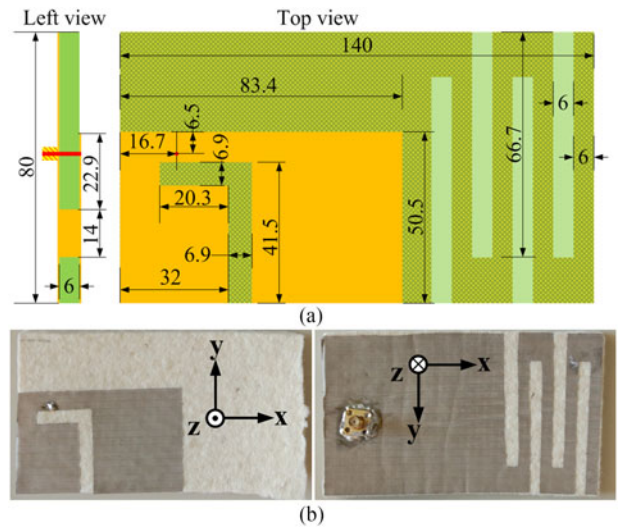

Fig. 1. Topology of the proposed antenna. (a) Dimensions in millimeters. The four slots in the ground have the same width, length, and spacing. (b) Prototype.

II. Antenna Topology and Operating Mechanism

The topology of the proposed antenna is shown in Fig. 1(a). A typical PIFA is used to cover the 433-MHz band. Several slots are introduced in the ground to extend the bandwidth. An L-shaped slot is introduced in the patch to achieve operation in the 2.4-GHz band.

A layer of 6-mm-thick felt is used as the substrate (relative permittivity is 1.3 and loss tangent is 0.044). The metallic layer is fabricated from a 0.17-mm-thick conductive textile sheet with a conductivity of 1.18×10^5 S/m and with a layer of thermal glue at its back. The patch, the shorting wall, and the ground are first cut by hand from a single piece of textile sheet and then adhered to the felt by heating with an iron. Finally, a standard SMA connector is soldered for measurement purposes. Soldering is done at lower temperatures to avoid damaging the materials. A photograph of the prototype is shown in Fig. 1(b).

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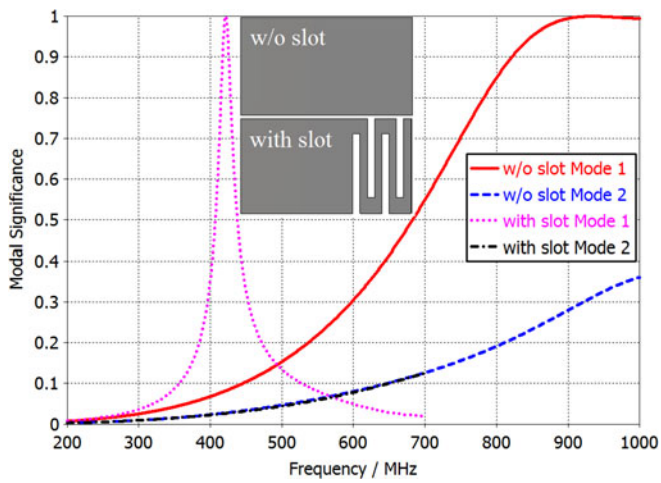

Fig. 2. Modal significance of a PEC sheet with/without slot.

The first key feature of the design is to increase the bandwidth around 433 MHz by exciting the ground plane. However, in order to allow ground planes much smaller than half a wavelength at 433 MHz, *y*-directional slots are cut into the ground, which extends the current path. This results in a ground length of only 140 mm. The characteristic modes [13] of a perfect electrical conductor (PEC) sheet with and without slots is analyzed with CST, and the modal significances are shown in Fig. 2. It can be clearly observed that the resonant frequency of the first characteristic mode of the ground plane is reduced from 934 to 427 MHz by introducing the slots. The reason is that the current of the first mode is flowing in the *x*-direction. The path of the current can be extended by the slots in the *y*-direction. Contrarily, the second mode, which has the current flowing in the *y*-direction, is almost not influenced by the slots [12]. It should be mentioned that the ground can be further miniaturized by cutting more slots, but at the cost of decreasing the matching bandwidth. A slotted PIFA is located on the edge exciting the ground. The distance from the short wall to the slot is 32 mm in the *x*-direction, which is about a quarter wavelength at 2.45 GHz. The exact location of the feeding port was obtained by optimizing the matching in both frequency bands.

III. Performance in Free Space

Fig. 3 shows the simulated and measured reflection coefficients in free space. An excellent agreement is achieved in both the lower and the upper bands. The lower matching bandwidths are 45 MHz (simulated) and 35 MHz (measured), which are much wider than the ISM band targeted (from 433.05 to 434.79 MHz). This wide bandwidth is obtained thanks to the radiation from the ground plane. A normal PIFA hardly achieves such wide bands with a compact topology. The higher matching bandwidths are 308 MHz (simulated) and 309 MHz (measured), which are also much wider than the targeted ISM band (from 2.4 to 2.5 GHz). A topology without slots in the ground was also simulated and compared in Fig. 3. It can be clearly observed that the matching is much worse in the lower band and that the upper band is almost not affected by the ground slots.

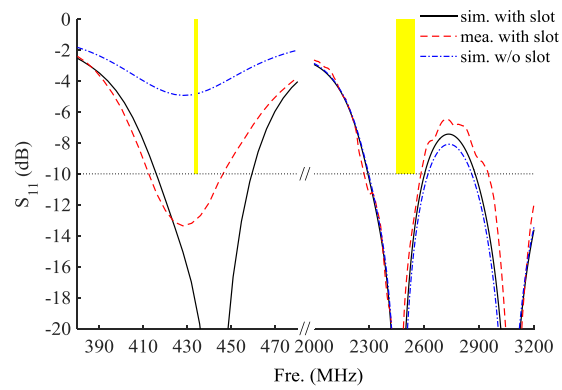

Fig. 3. Measured reflection coefficient. The yellow rectangles indicate the targeted frequency bands.

The far-field results are shown in Fig. 4. A very good agreement is achieved between simulated and measured results for the co-polar patterns in the *x*-*z* and *y*-*z* planes. The agreement is worse for the cross-polar components at the lower frequency, which may be due to the coupling between the antenna and the feeding cables. In the lower band, the radiation is mainly due to the first mode of the ground. Thus, an omnidirectional pattern is observed, similar as for a dipole. In the upper band, the radiation is mainly from the PIFA. Hence, the antenna is more directional. The gain and efficiency of the antenna are given in Table I.

IV. Performance on Human Body

The antenna was also characterized in on-body situations. Three different types of human phantoms were chosen in the simulation to mimic chest, shoulder, and arm. The chest model is a three-layered cubic model, including a 3-mm-thick layer of skin, a 7-mm-thick layer of fat, and a 60-mm-thick layer of muscle [17]. The total size of the phantom is $500 \times 500 \times 70$ mm^3. The shoulder model has the same layered structure as the chest model, but the size is $200 \times 100 \times 70$ mm^3. The arm model is a four-layered cylindrical model, including a 2-mm-thick layer of skin, a 5-mm-thick layer of fat, a 20-mm-thick layer of muscle, and a core of bone with radius 13 mm [17]. The total length of the cylindrical model is 350 mm. The parameters of all the tissues are obtained from the material library in CST. The antenna is located 10 mm above the surface of the skin, to mimic the separation by layers of clothes. For the arm model, the antenna is bent along two directions, respectively.

The simulated and measured reflection coefficients are shown in Fig. 5(a) and (b), respectively. The most noticeable conclusion is that the upper band (the 2.4-GHz band) is almost not influenced by the human body, even under the bending conditions. The reason is that the main radiating structure is the upper patch of the antenna. The ground plane shields the patch, in this way reducing the coupling between the upper patch and the human body. In contrast, the lower band (433 MHz) is more sensitive to the presence of the human body, since the ground of the PIFA is also contributing to the radiation, which will directly result in coupling with the human tissue. However, both the simulated and measured S_{11} are still below -6 dB in most situations,

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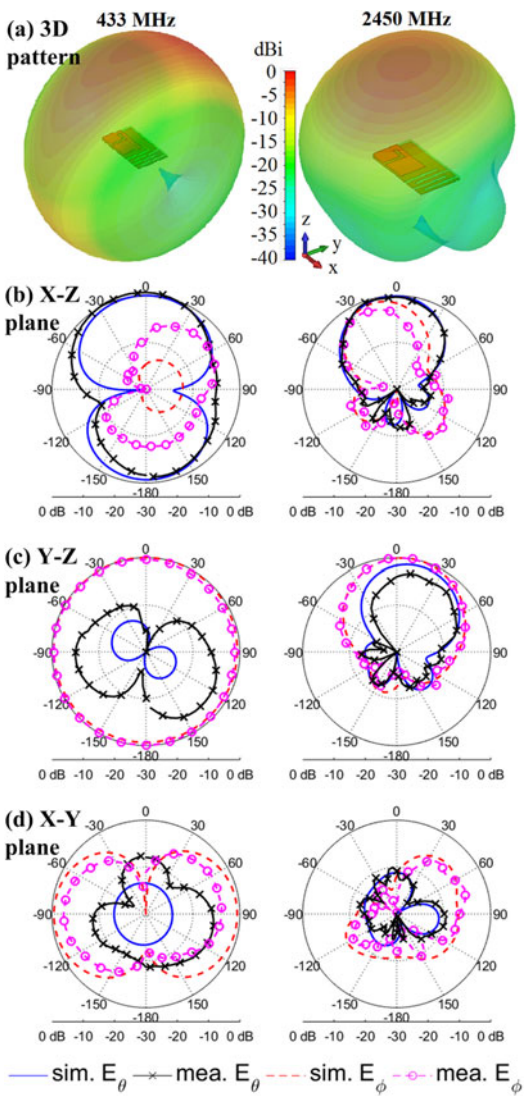

Fig. 4. Measured radiation patterns at 433 MHz (left) and 2.45 GHz (right). (a) Simulated three-dimensional (3-D) pattern, (b) simulated and measured patterns in the *x*-*z* plane, (c) simulated and measured patterns in the *y*-*z* plane, and (d) simulated and measured patterns in the *x*-*y* plane.

TABLE I
SUMMARY OF THE ANTENNA PERFORMANCE

		f_c (MHz)	BW (MHz)	eff. (%)	$G_{\text{real.}}$ (dBi)
Lower band	Sim.	437	45	48	−0.9
	Mea.	430	35	/	−0.5
Upper band	Sim.	2449	308	64	6.8
	Mea.	2428	309	/	6.1

which is still acceptable for most mobile applications. The only exception is in case of bending along the *x*-axis on the arm. The bending direction is then the same as the direction of the current flow on the ground, which may significantly change the impedance of the antenna.

The far-field performance was also verified numerically. In the upper band, since the radiated patch is shielded by the ground and the radiation has a broadside pattern, the on-body pattern

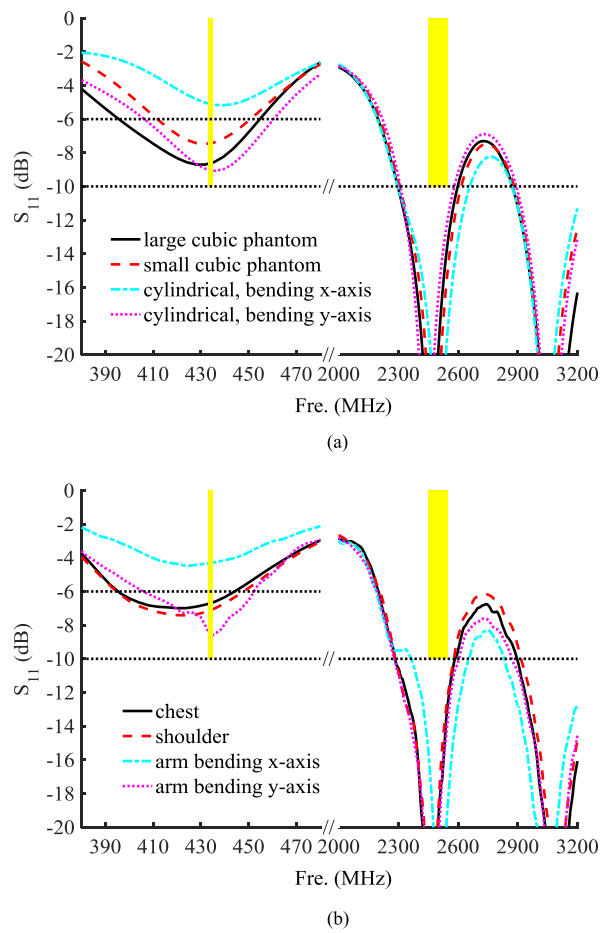

Fig. 5. Reflection coefficient on human body. (a) Simulated results and (b) measured results. The yellow rectangles indicate the targeted frequency bands.

is nearly the same as in free space, for all on-body configurations. The radiation efficiency decreases only about 10% since the back radiation is absorbed by the human body. In the lower band, the pattern of the antenna changes from a broadside to an omnidirectional one. The radiation efficiency drops sharply close to the human body. The simulated values are 7%, 8%, 6%, and 13% for the on-chest, on-shoulder, bending along the *x*-axis on arm, and bending along the *y*-axis on arm situations, respectively. A general conclusion is the following: The smaller the phantom present behind the antenna, the higher the efficiency. Thus, the ideal locations of the proposed antenna are on the arm (bending along the *y*-axis) or on the shoulder.

It can be pointed out that simulated and measured results do reach a relatively good agreement for this type of wearable antenna fabricated with conductive textile materials [18], [19].

V. CONCLUSION

A dual-band PIFA was designed. By co-designing the ground and the PIFA, a wide band (10%) is achieved around 433 MHz with a compact structure $(0.202 \times 0.115 \times 0.01\,\lambda_0^3)$. A slot is introduced in the patch to obtain an upper operating band around 2.4 GHz. A prototype was fabricated and measured. The measured results agree reasonably well with the simulated ones.

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Bending configurations and on body performance were also analyzed, illustrating the robust performance. The best locations are on the shoulder or on the arm (with bending along the y-axis). The simple topology of the proposed antenna is very suitable for large-scale industrial production.

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